

what does MP stand for?

west side park ♥



CS 340

Review and Final
Project

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Q1

~Code~
340



Updates

1. MP 10 due 5/5
2. Exam 3 on 4/30 ✓
 - a. Study guide and practice exam out!
3. 2nd chance exam next week Wednesday
4. Everything is graded! Please file regrade requests.
5. We have class Thursday (even with the exam).
 - a. Thursday - Luther Tychonievich guest speaker (with extra credit).
Ask him anything about systems!
6. No class Tuesday (per the student code)
7. Final Project Check-off May 11th 1:30 - 4:30

Agenda

1. Exam 3 Review
 - a. Aiohttp server
 - b. Authentication
 2. Final Project
 - a. MP 10
 - b. Final Project
 - c. Check-off
 3. Tetris AI
- 

AIOHTTP Server - reading a request

```
7 @routes.post("/guess")
8 async def post_func(req: Request) -> Response:
9     txt = await req.text()
```

mem

```
7 @routes.post("/guess")
8 async def post_func(req: Request) -> Response:
9     dict = await req.json()
```

Mem

```
79 multi = await req.multipart()
80 async for part in multi:
81     if part.name == 'png':
82         filedata = await part.read()
83     elif part.name == 'location':
84         city = await part.read().decode("utf-8")
```

~~mem~~

mem

you do not need to mem

AIOHTTP Server - calling another server

```
points_url = f"https://api.weather.gov/points/{lat},{lon}"  
async with aiohttp.ClientSession() as session:  
    async with session.get(points_url) as resp:  
        point_data = await resp.json()
```

↑ .text

~~mem~~

AIOHTTP Server - sending a response

```
11 @routes.post("/incr")
12 async def incr(req: Request) -> Response:
13     global state
14     state += 1
15     return Response(text=str(state), status=200)
```

Handwritten annotations for the first code block:

- A blue circle around the word `mem` in the top right corner.
- A red arrow pointing down to the `state += 1` line.
- A red circle around the `text=str(state)` argument.
- A red circle around the `status=200` argument, with the text `status=200` written in red above it.

```
28 @routes.get("/users")
29 async def get_users(req: Request) -> Response:
30     return json_response(list(users.keys()), status=200)
```

Handwritten annotations for the second code block:

- A blue circle around the word `mem` in the top right corner.
- A red circle around the `list(users.keys())` argument, with the text `list dict` written in red below it.
- A red circle around the `status=200` argument, with the text `status=200` written in red above it.

What is used to safely store a password in a web server database?

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Q2

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A. Salting

— adding random string to the password

B. Hashing

— 1-way

C. Encryption

— 2-way with a key

D. Rainbow Table

— pre computed password hashes

E. Diffie-Hellman

— a way to exchange a key

user1

password1

AXB76

Which are(is) correct?

- ☒ A. Encryption can be reversed with a key
- ☐ B. Hashing can be reversed with a key
- ☒ C. Encryption is often used for sending data on the internet
- ☐ D. Hashing is often used for sending data on the internet

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Q3

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Agenda

1. Exam 3 Review
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2. Final Project

- a. MP 10**
- b. Final Project**
- c. Check-off**

3. Tetris AI

MP 10 - Tetris Server

server tetris.py → send board updates to client

client index.html → send key presses

web sockets

many
clients

func websocket_handler,
get a tetris board

⌚ waiting for
key presses

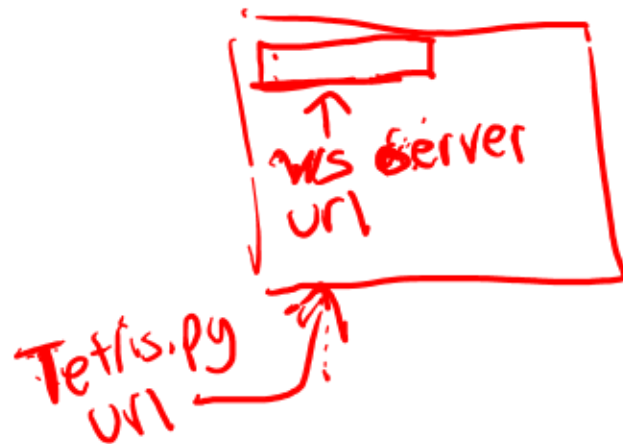
Final Project - **bot.py**

tetris.py ← server ← sends board state

bot.py ← client
↑ sends key presses

add/watch

bot.py serves
botui.html



Final Project - Check-off [Flatris]

I will run flatris.py server

← instead of
tetris.py

↓
pair off
bots and people
for games

Competitive Tetris

Run your bot
play tetris
vote

Run your bot

~~open~~ run in your VM
bot.py

open botui.html
connect to my server

Play flatris

Vote

- most human like bot
- best spotter

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Is _____ intelligent?

- A. An ant
- B. A tree
- C. An average human
- D. An average computer
- E. A calculator
- F. MP 9 Weather Filter

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~~Artificial~~ Intelligence

- learning $\leftarrow$ changing behavior based on previous experience
- adaptability
- logical reasoning
- analogies $\leftarrow$ transfer learning

Brief History of AI

1950 - Alan Turing paper “Computing Machinery and Intelligence”

1955 - The term AI was introduced

1959 - Concept of machine learning developed

1997 - IBM defeats world champion at chess

2016 - Google Deepmind beat world champion at go

2020 - OpenAI introduce GPT-3

AI Terms

AI approaches

ML
machine learning
supervised
unsupervised
reinforcement learning
classical / symbolic

AI techniques

deep learning (neural network), decision trees

Applications

Robotics, NLP, computer vision,

The Turing Test

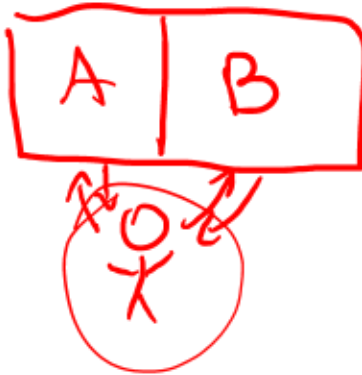
Can machines think?

← Big question



→ smaller

Can a machine exhibit intelligent behavior indistinguishable from that of a human?



Given the turing test, what percentage of the time should the participant be able to correctly identify the human?

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50%



Prof. Schatz Turing Test

What do you love about teaching CS 340?

A) AI 27%

"CS 340 is where students finally get to see why their code works, not just that it does. There's something really special about watching someone go from intimidated by low-level programming to genuinely excited that they can see all the way down to the hardware."

Pick human!
B) which one did I write?

"I love teaching CS 340 for many reasons. I think the content is interesting and at a level that is intuitive to understand. This allows the course to cover a lot of different topics during only 1 semester."

Human 73%

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Turing Test History

1950 - Alan Turing introduced the idea

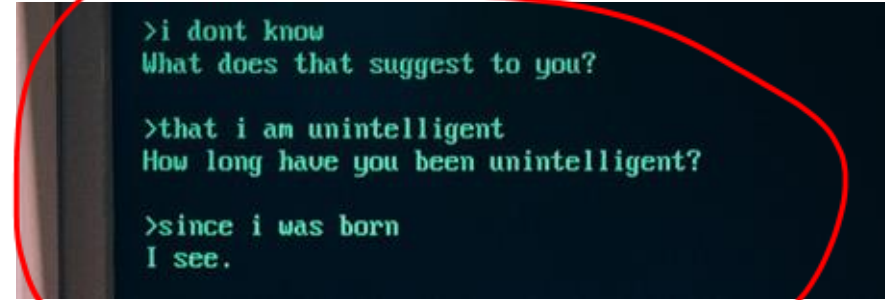
1996 - Weizenbaum “succeeded”

2001 - Goostman “succeeded”

2025 - A study published evaluating GPT on
the turing test



GPT-4.5 was identified
human 73% of the time



**We passed the turing test...
but have we achieved
artificial general intelligence?**

20% yes

50% no but almost

29% no and we never will

1% unsure

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Q7

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CAPTCHA - What does it stand for?

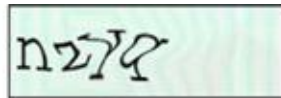
completely automated Public Turing test to tell Computers and Humans apart



Gimpy



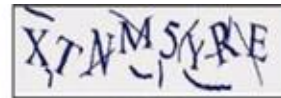
EZ Gimpy



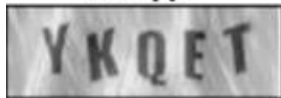
Gimpy-r



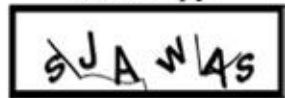
Mailblocks



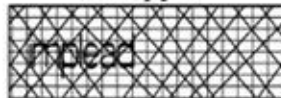
MSN



Register



Yahoo



Ticketmaster



Google



Holiday Inn



BotBlock



BotCheck



Megaupload



LinkedIn



Cryptograph



Chinese CAPTCHA



Baidu



3D CAPTCHA



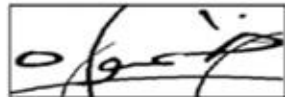
STE3D-CAP



DevaCAPTCHA



Gurmukhi
CAPTCHA



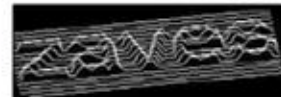
Arabic CAPTCHA



BotDetect



TeaBag CAPTCHA
1.2



Super CAPTCHA

Cognitive Modeling

models are inspired
by the human mind



Tetris Turing Test (Final Project)

Trick your classmates

What are some strategies for playing tetris?

Simple - find lowest spot

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Tetris Research

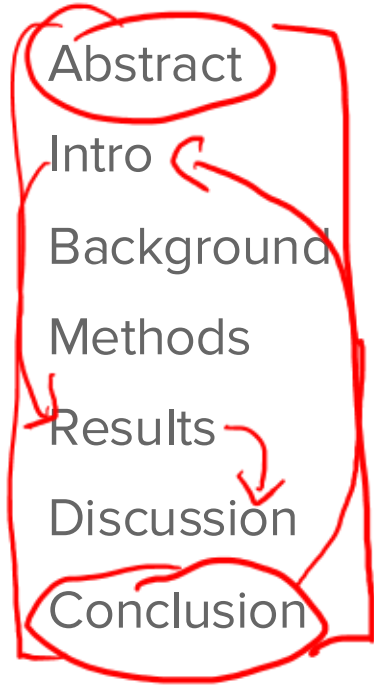
Sibert, Gray, and Lindstedt 2015 explores human performance against an AI model.

Böhem, Kókai, and Mandl 2005 list 12 heuristics and a learning approach to picking how to combine them.

Thiery and Scherrer 2009 contains a table of 32 heuristics and describes how to combine six of them to implement Dallacherie's unpublished but very effective approach from 2003.

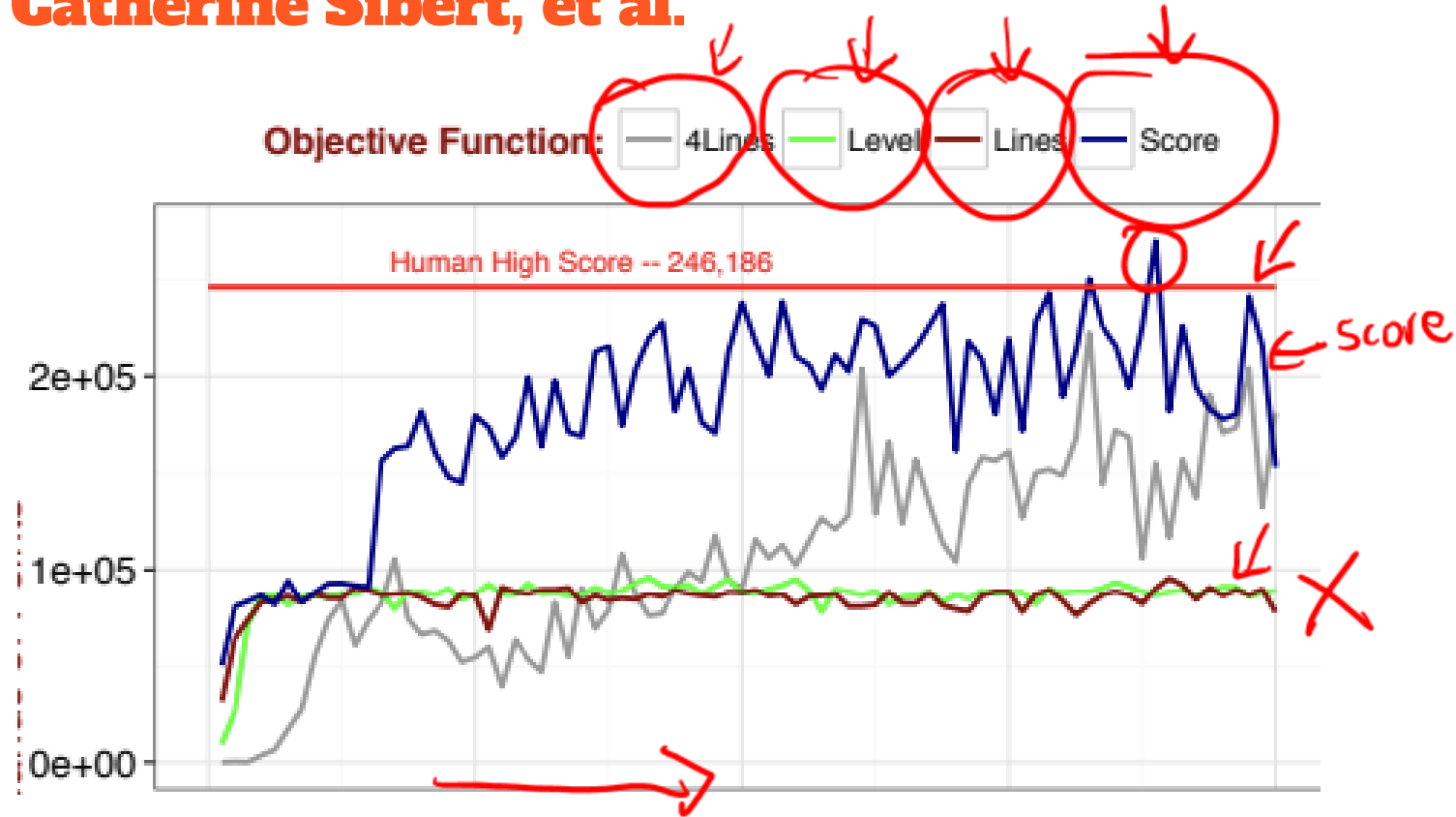
Bergmark 2015 describe a heuristic based on identifying three types of holes in detail, with suggestions on how to score and pick moves.

How to read a research paper



skip around

Tetris™: Exploring Human Performance via Cross Entropy Reinforcement Learning Models (2015) - Catherine Sibert, et al.



Prediction: Do you think the Tetris Model is better at predicting expert players or novice players?

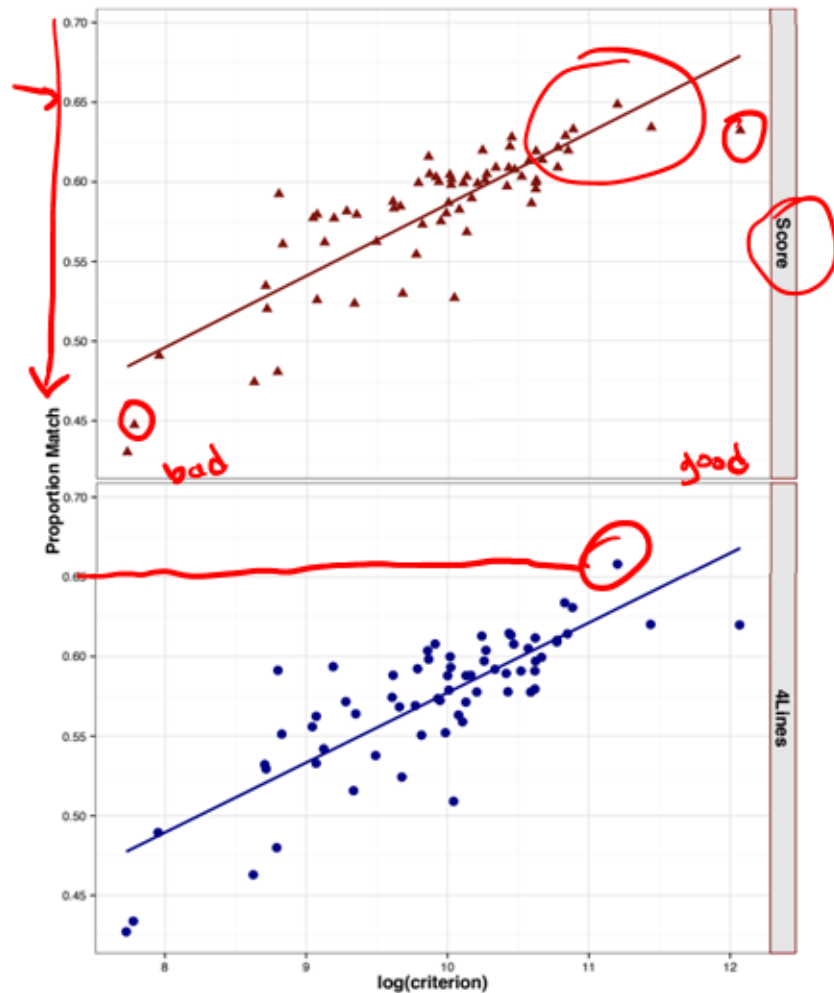
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Predicting Humans



Building Controllers for Tetris - Christophe Thiery, Bruno Scherrer

Feature	Description
Max height	Maximum height of a column
Holes	Number of empty cells covered by a full cell
Column height	Height of each column
Column difference	Height difference between each pair of adjacent columns
Landing height	Height where the last piece is added
Cell transitions	Number of full to empty or empty to full cell transitions
Deep wells	Sum of well depths, except for wells with depth 1
Embedded holes	Sort of weighted sum of holes (not precisely documented)
Height differences	Sum of the height differences between adjacent columns
Mean height	Mean height of columns
Δ max height	Variation of the maximum column height
Δ holes	Variation of the hole number
Δ height differences	Variation of the sum of height differences
Δ mean height	Variation of the mean column height
Removed lines	Number of lines completed at the last move
Height weighted cells	Full cells weighted by their height
Wells	Sum of the depth of the wells
Full cells	Number of occupied cells on the board
Eroded piece cells	(Number of rows eliminated in the last move) $\times$ (Number of bricks eliminated from the last piece added)
Row transitions	Number of horizontal cell transitions
Column transitions	Number of vertical cell transitions
Cumulative wells	$\sum_{w \in \text{wells}} (1 + 2 + \dots + \text{depth}(w))$
Min height	Minimum height of a column
Max – mean height	Maximum column height – Mean column height
Mean – min height	Mean column height – Minimum column height

Final Project - Tetris Bot

can't be
faster than
fastest human

Example — Starter Bot

When you get a message with `next` in it (i.e. a new tetromino is on the board), pick a full sequence of moves as follows:

1. Use your MP10 logic to loop through all possible horizontal positions and orientations of the new tetromino, finding where it would drop from each position.
2. Evaluate those options based on a heuristic to find the “best”; as a starter the “best” option maximizes the sum of the y coordinates of the cells in the tetromino.
3. Emit `left`, `right`, `cw`, and/or `ccw` commands to get the tetromino into that horizontal location and orientation, followed by `drop`.

Final Project - Expectations

- VM on
- latest code on your VM
- before - run bot.py and tetris.py on
the VM and watch your bot
play

FLEX Survey - Let me know what went well and what needs improvement!



[https://go.blueja.io/X7_x5QhS
xEi42EUOO7M5SA](https://go.blueja.io/X7_x5QhSx Ei42EUOO7M5SA)

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